# Technical Report

# AI-Powered Image Classification Platform

End-to-end deep-learning image classification using transfer learning, REST API, Streamlit, and prediction history.

## 1. Objective

Develop a production-oriented image classification platform that trains and compares multiple deep-learning models, evaluates them using standard classification metrics, exposes image prediction through a REST API, and provides a simple browser-based interface.

## 2. Dataset

CIFAR-10 is used as the public dataset. It contains 60,000 32×32 RGB images across ten classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck.

Official source: https://www.cs.toronto.edu/~kriz/cifar.html

## 3. Data Pipeline

```
CIFAR-10
   |
   +--> Train / Validation / Test split
   |
   +--> Resize to 224x224
   |
   +--> Training augmentation
   |       - Random horizontal flip
   |       - Random crop
   |       - Color jitter
   |
   +--> ImageNet normalization
   |
   +--> Transfer learning
          |-- ResNet18
          `-- MobileNetV3-Small
```

The demonstrated training run used 10,000 training images and 2,000 validation images. Final evaluation was performed on the full 10,000-image CIFAR-10 test set.

## 4. Model Architecture and Training Strategy

**ResNet18:** ImageNet-pretrained residual network with a new ten-class classification layer.

**MobileNetV3-Small:** ImageNet-pretrained lightweight convolutional network with a new ten-class classifier.

For the demonstrated run, the convolutional backbone was frozen and the classification head was fine-tuned using AdamW with learning rate 1e-3, weight decay 1e-4, and 5 epochs. Training ran on an NVIDIA GeForce GTX 1650 using CUDA.

## 5. Performance Comparison

| Model | Best validation accuracy | Test accuracy (2k demo subset) |
|---|---|---|
| ResNet18 | 78.10% | — |
| MobileNetV3-Small | 81.45% | 80.65% |

MobileNetV3-Small was selected because it achieved the strongest validation accuracy in the demonstrated run. Its final model was then evaluated on the full CIFAR-10 test set.

## 6. Final Full-Test Evaluation

| Metric | Result |
|---|---|
| Accuracy | 80.93% |
| Precision (weighted) | 81.74% |
| Recall (weighted) | 80.93% |
| F1-score (weighted) | 80.86% |
| Top-5 accuracy | 99.28% |

The confusion matrix is generated by the evaluation pipeline and stored as reports/confusion_matrix.png.

## 7. Deployment Architecture

```
User
  |
  v
Streamlit Web UI
  |
  v
FastAPI REST API
  |
  +--> PyTorch MobileNetV3-Small
  |
  +--> SQLite prediction history
  |
  `--> JSON: top-1 + top-5 confidence scores
```

The REST service exposes /health, /classes, /history, and /predict. The Streamlit frontend provides image upload, classification, confidence scores, and prediction history.

## 8. Demonstrated Prediction

A CIFAR-10 test image with ground-truth class **cat** was uploaded through the application. The model correctly predicted **cat** with **85.5%** confidence. The top-five output also displayed dog, bird, frog, and airplane with lower confidence.

## 9. Validation and Testing

Automated API tests were executed with pytest: **2 passed**. Four console warnings were deprecation warnings from the FastAPI/Starlette testing stack and did not cause failures.

## 10. Deliverables

Source code, training scripts, experiment notes, trained PyTorch model, evaluation artifacts, REST API, Streamlit frontend, Docker configuration, README documentation, and technical report are included in the project package.

## 11. Conclusion

The platform demonstrates the complete workflow from public data ingestion and augmentation through transfer learning, model comparison, quantitative evaluation, REST deployment, browser-based inference, and persistent prediction history. MobileNetV3-Small achieved 80.93% accuracy and 99.28% top-5 accuracy on the full CIFAR-10 test set for the demonstrated training configuration.